

EXPERTS AND EXPERTISE IN COLONIAL AFRICA RECONSIDERED: SCIENCE AND THE INTERPENETRATION OF KNOWLEDGE

WILLIAM BEINART, KAREN BROWN, AND DANIEL GILFOYLE

ABSTRACT

Africanists have long criticized the social construction, and consequences, of technical knowledge. Colonial science was seen as a particularly problematic enterprise, moulded by authoritarian colonial states, wherein science ‘delineated the relationship of power and authority between rulers and ruled’. Much the same critique has been applied to post-colonial experts and expertise, becoming almost paradigmatic in the literature. This article seeks to re-open this debate, pointing to the diverse and changing location of scientists; the salience of scientific work in constructing categories and understandings for historians and social scientists; the value of trying to understand scientific explanations, as opposed simply to analyse their application in coercive policies; and the degree to which experts have sometimes incorporated local knowledge. The article draws examples from veterinary science and policy in southern Africa, and seeks to move beyond the inversions of colonial thinking in post-colonial analysis and provide instead a platform for interdisciplinary research strategies.

COLONIAL SCIENTISTS AND TECHNICAL OFFICERS IN AFRICA have received a bad academic press. This article attempts to expand the analysis of science in Africa. These issues are too broad to cover systematically in a single article; we use examples drawn largely from southern Africa, including our research on the history of environmental and especially veterinary sciences between about 1870 and 1960. Veterinary science was particularly significant in South Africa, where livestock were so central to production. Veterinary research and regulation probably absorbed more than half the colonial and national agricultural budgets in the period from the appointment of the first government vets in the 1870s to the 1930s.

William Beinart is Rhodes Professor of Race Relations at Oxford University (william.beinart@sant.ox.ac.uk); Karen Brown is a senior research officer at the Wellcome Unit for the History of Medicine, Oxford (karen.brown@wuhmo.ox.ac.uk); and Daniel Gilfoyle works in Research and Collections at the National Archives in London (Danielgilfoyle@nationalarchives.gsi.gov.uk). We acknowledge funding from the Wellcome Trust for a joint project on ‘Veterinary medicine, entomology and the state in South Africa, c. 1900–1950’, based at the Wellcome Unit for the History of Medicine, University of Oxford, 2002–5.

Our focus is historical but we believe that a debate about this period, extensively excavated in recent research, has relevance to more contemporary issues. We are suggesting a more open curiosity about colonial and post-colonial science and are seeking routes by which to move beyond the intellectual inversions so central to anti-colonial and post-colonial analyses. We are also interested in pursuing other strands of discussion such as the essential place of scientific approaches in understanding the history of disease and environment in Africa, and establishing a more secure basis for interdisciplinary research. Interventions such as this article, concentrating on a specific theme, tend to be read as polarizing debates. That is not our aim. We accept the value of the rich literature on the history of technical mishaps and coercive colonial intervention. But we suggest that this pattern of analysis obscures important and interesting developments in science and 'expertise' in Africa.

From their foundations, in the dying days of colonialism, African history and social sciences self-consciously tried to write from the vantage point of Africans and to decolonize European minds. By the late 1970s many academics were disillusioned with nationalist narratives in the light of the corruption of their apparent aims, fifteen to twenty years after independence. There was widespread unease with easy assumptions about modernization and development, so closely linked to both the late-colonial and nationalist projects. This included a critique of socialist policies in Tanzania, Mozambique, and elsewhere, especially in their ambitious plans to transform rural societies. Academic work focused more on rural communities, on chieftaincy or traditional authority, and on women as the most oppressed category of people in Africa. It concentrated on ethnicity as much as nationalism, and on cultural continuities as much as social change. Continuities were also detected in the colonial and post-colonial state: both were seen as authoritarian and as disadvantaging rural society.

Such perspectives, 'taking the part of peasants', and drawing on strong anthropological traditions, as well as social history 'from below', underpinned an expanding body of literature on development, agrarian issues, environmental change and rural resistance.¹ Terence Ranger on *Peasant Consciousness* in Zimbabwe and Paul Richards on *Indigenous Agricultural Revolution* in Sierra Leone were important exemplars.² Some of this literature is reviewed in an *African Affairs* article in 2000 and we will not revisit

1. Gavin Williams, 'Taking the part of peasants: rural development in Nigeria and Tanzania' in P. C. W. Gutkind and I. Wallerstein (eds), *The Political Economy of Contemporary Africa* (Sage, Beverly Hills, CA, 1976), pp. 131–54; William Beinart and Colin Bundy, *Hidden Struggles in Rural South Africa* (Ravan Press, Johannesburg, 1987).

2. Terence Ranger, *Peasant Consciousness and Guerrilla War in Zimbabwe: A comparative perspective* (James Currey, London, 1985); Paul Richards, *Indigenous Agricultural Revolution: Ecology and food production in West Africa* (Hutchinson, London, 1985).

it in detail here.³ Protagonists explored and advocated the importance of rural initiative and participation in the planning and process of development. Ideas about agro-forestry and community or local management of natural resources came to the fore in environmental studies. Research burgeoned on indigenous or local knowledge, and, as a corollary, its counterpoint, the critique of colonial science. Academics (here Richards was to an extent an exception) criticized science in the service of Western (sometimes male) power as a whole, but colonial science was seen as a particularly problematic enterprise, moulded by authoritarian colonial states. Science, David Arnold argued, 'delineated the relationship of power and authority between rulers and ruled'.⁴ Race, as well as knowledge systems and power, was part of this asymmetrical set of relations between colonizers and colonized. Africanists insisted on specifying the social location of scientists and technical officers and, in the growing field of medical history, on seeing scientific and medical knowledge as socially constructed.⁵ 'Western biomedicine', Shula Marks noted, 'has undoubtedly played a major role both in making universalizing claims, and in creating and reproducing racial and gendered discourses of difference'.⁶ By no means all this literature shared the same ideological and methodological approaches. We will focus briefly on an article by John McCracken, 'Experts and expertise in colonial Malawi', published in this journal in 1982.⁷ He succinctly summarized many key points and identified interesting and specific examples.

McCracken emphasized that experts were deeply constrained by their social location in the colonial system. Cotton-growing experts in Malawi, for example, certainly believed that Africans could be more effective export commodity producers than settlers and, in the context of southern Africa, this was a position unusually sympathetic to the peasantry. But 'good intentions were often fatally flawed by the authoritarian and paternalistic attitudes of its staff'. He echoed the frequently made point that colonial experts were

3. William Beinart, 'African history and environmental history', *African Affairs* 99, 395 (2000), pp. 269–302.

4. David Arnold, 'Introduction', in D. Arnold (ed.), *Imperial Medicine and Indigenous Societies* (Manchester University Press, Manchester, 1988), p. 2.

5. Randall M. Packard, *White Plague, Black Labour: Tuberculosis and the political economy of health and disease in South Africa* (University of Natal Press, Pietermaritzburg, 1989); Megan Vaughan, *Curing Their Ills: Colonial power and African illness* (Stanford University Press, Stanford, CA, 1991). Maryinez Lyons, *The Colonial Disease: A social history of sleeping sickness in Northern Zaire, 1900–1940* (Cambridge University Press, Cambridge, 1992). For historiographical developments more generally, Mark Harrison, *Disease and the Modern World, 1550 to the Present Day* (Polity, Cambridge, 2005).

6. Shula Marks, 'What is colonial about colonial medicine? And what has happened to imperialism and health?', *Social History of Medicine* 10, 2 (1997), pp. 205–19.

7. John McCracken, 'Experts and expertise in colonial Malawi', *African Affairs* 81, 322 (1982), pp. 101–16. A version of this article was first given at a conference to mark Professor John McCracken's retirement ('Malawi after Banda', Centre for Commonwealth Studies, Stirling University, 4–5 September 2002).

often out of their depth. They had 'limited understanding of the complex and changing nature of ecological conditions in the Lower Shire valley', where cotton had been adopted. They were inconsistent, at one time advocating pure stands, then mixed cropping.⁸ Malawi's cotton was beset with the bollworm, and experts were equally inconsistent in trying to enforce the best planting times to control this pest. This argument about inadequate research, especially before adopting policies with major implications for the peasantry, is central to the literature.

Experts also proved unable to deal effectively with the spread of tsetse fly and trypanosomosis. The basic causes of the disease were known by the early twentieth century, but control was elusive. Officials first experimented in catching flies, sometimes using human traps.⁹ Mechanical trapping devices were also tried as well as extensive game culling and bush clearance in order to remove the habitat that favoured the fly. The shooting of wildlife had the effect of scattering animals, and probably the disease, over a wider area. Another strategy adopted in Malawi, as in other parts of colonial Africa in the early decades of the twentieth century, required concentration of settlements. This was socially disruptive, difficult to administer, and of doubtful effectiveness.

Colonial experts in Malawi, McCracken argued, were captured by the soil erosion mania, especially in the 1930s and 1940s. Massive contouring campaigns were coercive, ignored well-established local methods of cultivation, and were met with intense resistance. This was one of the key agrarian interventions by the colonial state in Malawi. He also illustrated the inadequacy of measures to combat East Coast fever, a tick-borne disease of cattle, by dipping. In this case, it was not so much the imposition of a scientific strategy that was the subject of critique as the colonial administration's failure to introduce dipping tanks in the north, where most African cattle were kept. There is a tension in the literature as to whether colonial states did too much or too little.

Experts, McCracken contended, somewhat eliding the colonial and post-colonial periods, swooped in, dispensed superficial solutions, often hopelessly inappropriate, and then moved on; they seldom had to take responsibility for their recommendations. They were dry-season travellers sticking closely to the road – the 'worst plague of locusts ever to descend on the poor countries'.¹⁰ McCracken emphasized the growing importance of colonial technical officers and scientists in Malawi from the interwar years, but illustrated the chasms in their understanding of the African

8. McCracken, 'Experts and expertise', pp. 102–5.

9. See James Busvine, *Disease Transmission by Insects: Its discovery and ninety years of efforts to prevent it* (Springer Verlag, Berlin, 1993), pp. 168–9 for this strategy on Principe in the Gulf of Guinea.

10. McCracken, 'Experts and expertise', p. 116, quoting Jean Houbert.

environment, their impotence, and in some cases their hubris. While he briefly doubted that 'practical experience in itself can provide an adequate substitute for scientific knowledge', his general analysis captured wonderfully the mood of the time: 'the best experts are those who recognize that rural people are the best experts'.¹¹ Here and elsewhere, we are left with the impression that experts could, and can, do little for the people, except learn from them.

Scholars, including Beinart, were reading through and behind colonial archives – as African historians often did – seeking rural and peasant voices, in order to explore colonial mishaps as well as African consciousness.¹² As the critique of post-colonial development gathered strength, it was important to demonstrate the more coercive elements of agrarian policy and its failures. Such perspectives were powerfully developed in overviews such as Michael Adas's *Machines as the Measure of Men* and perhaps reached their Africanist apogee in Melissa Leach and James Fairhead's *Misreading the African Landscape*.¹³ James Scott's *Seeing like a State* epitomized, on a broader front, the anti-statist critique of 'high modernism' and scientific planning.¹⁴ Quoting Isaiah Berlin, Scott compared the 'the scientific forester', the 'cadastral officer', the administrative man, to the hedgehog, 'who knows only one big thing', while farmers, peasants and naturalists were like the fox 'who knows a great many things'.¹⁵ Hedgehogism narrows knowledge, and provides a 'rather static and myopic view'. Administrative man, the utilizer of science, 'perceives a drastically simplified model of the buzzing, blooming confusion that constitutes the real world'.¹⁶

While it is clearly mistaken to overestimate the impact of academic work, such analyses were part of a far-reaching reinterpretation of science, knowledge, and environmental management. They helped to change the terms of debate and policy – though many would say not yet sufficiently so. By the mid-1990s this was the new paradigm in African Studies. It shared some features of broader post-modernist approaches, although Africanist research preceded the latter, and often retained its own distinctive, empirical, and fieldwork-based qualities. African environmental history and social sciences were booming, and these ideas underpinned conferences and richly

11. *Ibid.*, p. 116.

12. William Beinart, 'Soil erosion, conservationism, and ideas about development: a southern African exploration', *Journal of Southern African Studies* 11, 1 (1984), pp. 52–83; Beinart and Bundy, *Hidden Struggles in Rural South Africa*.

13. Michael Adas, *Machines as the Measure of Men: Science, technology, and ideologies of Western dominance* (Cornell University Press, Ithaca, NY, 1989); James Fairhead and Melissa Leach, *Misreading the African Landscape: Society and ecology in a forest-savanna mosaic* (Cambridge University Press, Cambridge, 1996).

14. James Scott, *Seeing Like a State: How certain schemes to improve the human condition have failed* (Yale University Press, New Haven, CT, 1998).

15. *Ibid.*, p. 45.

16. *Ibid.*, p. 45.

researched books from Elias Mandala's *Work and Control in a Peasant Economy* to Dan Brockington's *Fortress Conservation*.¹⁷

Our concerns about the limits of this approach developed from our research projects in the 1990s, focusing especially on South Africa, as well as Ravi Rajan's work on Indian foresters, and Helen Tilley's on the scientists involved in Lord Hailey's *African Survey*.¹⁸ Scientific and technological developments were, as Daniel Headrick argued, at the heart of empire.¹⁹ Especially from the mid-nineteenth century, scientists were central, if not always direct, actors in imperial development. They helped to pioneer new technologies that facilitated vastly more effective exploitation of natural resources, for agriculture and industry. Technology underpinned growing superiority in European armaments and communications. Constraints imposed by environment and disease were gradually driven back. The scale of European imperialism and its transformative capacities were shaped and facilitated by science, technology, and environmental transformation.

Science was intrinsic to the processes of imperialism and settler colonialism. Clearly there were branches of colonial interest, such as racial sciences, where it is difficult to separate these connections.²⁰ We want to ask, however, whether science and technology can be partially detached from, rather than

17. For example, Elias C. Mandala, *Work and Control in a Peasant Economy: A history of the lower Tchiri Valley in Malawi, 1859–1960* (University of Wisconsin Press, Madison, WI, 1990); Roderick Neumann, *Imposing Wilderness* (University of California Press, Berkeley, CA, 1998); Terence Ranger, *Voices from the Rocks: Nature, culture and history in the Matopos Hills of Zimbabwe* (James Currey, Oxford, 1999); Dan Brockington, *Fortress Conservation: The preservation of the Mkomazi Game Reserve, Tanzania* (James Currey, Oxford, 2002); K. A. Hoppe, *Lords of the Fly: Sleeping sickness control in British East Africa, 1900–1960* (Praeger, Westport, CT, 2003); Nancy Jacobs, *Environment, Power, and Injustice – A South African history* (Cambridge University Press, Cambridge, 2003). See also William Beinart (ed.), *The Politics of Conservation in Southern Africa*, special issue of *Journal of Southern African Studies*, 15, 2 (1989).

18. William Beinart, 'Vets, viruses and environmentalism: the Cape in the 1870s and 1880s', *Paideuma* 43 (1997), pp. 227–52; William Beinart, 'Men, science, travel and nature in the eighteenth and nineteenth century Cape', *Journal of Southern African Studies* 24, 4 (1998), pp. 775–99; William Beinart, *The Rise of Conservation in South Africa: Settlers, livestock, and the environment, 1770–1950* (Oxford University Press, Oxford, 2003); Daniel Gilfoyle, *Veterinary Science and Public Policy at the Cape, 1877–1910* (University of Oxford, unpublished DPhil thesis, 2002); Karen Brown, *Progressivism, Agriculture and Conservation in the Cape Colony, c. 1902–1908* (University of Oxford, unpublished DPhil thesis, 2002); Ravi Rajan, 'Imperial environmentalism or environmental imperialism? European forestry, colonial foresters and the agendas of forest management in British India, 1800–1900' in Richard Grove, Vinita Damodaran, and Satpal Sangwan (eds), *Nature and the Orient: The environmental history of South and South East Asia* (Oxford University Press, Delhi, 1998), pp. 324–71; Helen Tilley, *Africa as a 'Living Laboratory': The African Research Survey and British colonial empire – consolidating environmental, medical, and anthropological debates, 1920–1940* (University of Oxford, unpublished DPhil thesis, 2001).

19. Daniel R. Headrick, *The Tools of Empire: Technology and European imperialism in the nineteenth century* (Oxford University Press, Oxford, 1981) and *The Tentacles of Progress: Technology transfer in the age of imperialism, 1850–1940* (Oxford University Press, Oxford, 1988).

20. Saul Dubow, *Scientific Racism in South Africa* (Cambridge University Press, Cambridge, 1995).

collapsed into the immediate priorities of colonialism? How do we begin to think about the operation of scientific modes of rationality in the empire, multiple as they were, in a more complex framework? Should this affect how we read archives, choose our examples, and balance our findings?

Scientific officials and their experiences: colonial science and science in the colonies

McCracken, and many since, commented on the fleet-footedness of experts and their metropolitan location. Yet a number spent sustained periods of work in African or other colonized countries. Settler and colonial states, particularly South Africa and Zimbabwe, as well as India, created careers for them in a way that was unusual for social scientists. South Africa was able to generate its own scientific institutions, careers, and specialisms in a setting that was African at least in a physical sense. Most of the key South African officials involved in veterinary and environmental research from the late nineteenth century spent at least 25 years in government service.²¹ Both the South African and British colonial bureaucracies increasingly demanded scientifically trained officers for their technical services. In spheres such as soil conservation, environmental regulation, and especially control of animal diseases, scientific officials worked across the farmlands and African reserves. Russell Thornton, for example, started his career in the Cape and Union Departments of Agriculture, largely serving white farmers. In 1929 he became Director of Agriculture in the Native Affairs Department of South Africa, then moved in the 1930s to Lesotho where he supervised the anti-erosion drives, and advised on strategies elsewhere.

Technical officers and scientists often became part of a nationalist rather than imperial project; in India, particularly, this could involve opposing both British rule and the Gandhian influence in the Indian Nationalist Congress. Some believed, Dinesh Abrol notes, that science alone was 'capable of improving conditions of life when fully applied in a planned economy' and they rejected the 'dubious Gospel of the Spinning Wheel and the Bullock Cart'.²² Saul Dubow traces a similar assertiveness in South Africa, with Smuts in the vanguard.²³ Africans were, to a far greater degree than Indians, excluded from scientific training until the late colonial period. One of the earliest South African veterinary surgeons was Jotello Soga, son of the first African Presbyterian minister Tiyo Soga and his Scottish wife. He worked with both

21. Beinart, *Rise of Conservation*; Saul Dubow, *A Commonwealth of Knowledge: Science, sensibility, and white South Africa 1820–2000* (Oxford, Oxford University Press, 2006).

22. Dinesh Abrol, "'Colonized minds' or progressive nationalist scientists: the Science and Culture Group' in Roy MacLeod and Deepak Kumar (eds), *Technology and the Raj: Western technology and technical transfers to India, 1700–1947* (Sage Publications, New Delhi, 1995), p. 66.

23. Dubow, *A Commonwealth of Knowledge*.

white and black farmers in the late nineteenth- and early twentieth-century eastern Cape. But he proved to be a rare exception. South African scientists in government employment were very largely white up until the homeland era. Nevertheless, as in the South Asian literature, any analysis of interventions in Africa over the last half century has to include the growing numbers of African experts.

The issue of experience, time on site, and physical and intellectual location crops up in historical debates and recent literature. Nineteenth-century Cape farmers railed against the experts – who were trying to do something about their scabby sheep or their profligate burning of vegetation – for not being there long enough. In their view, scientific officials were not practical men, in the language of the time, but theoretical men. Afrikaner farmers advocated home-grown solutions for an apparently local disease of drought and environmental poverty. Competing patterns of knowledge, as well as questions concerning the legitimacy and authenticity of experts, swirl around these apparently straightforward arguments about years and experience. Our research on veterinary officers suggests that they often developed an in-depth knowledge, born of extended careers in colonial contexts. Duncan Hutcheon, veterinary surgeon at the Cape from 1881 to 1907, pioneered prophylaxes against a range of animal diseases, bridging black and white livestock owners. After Union (1910) Onderstepoort Veterinary Institute near Pretoria became a centre of bio-medical research for the region.²⁴ Its first two directors, the Swiss bacteriologist Arnold Theiler (1908–27), and the South African Petrus du Toit (1927–48) supervised extended research programmes afforded by their growing institutional base. It is probably not worth doing the research, but it is quite likely that colonial experts spent more time in Africa on average than Western-based social scientists and historians do now. There are institutional as well as financial reasons for this, and transport has become quicker and cheaper.

Some scientific officers did move regularly, but they nevertheless accumulated experience and knowledge of particular issues, and their careers demonstrated a professional continuity. Mobility, and experience in different areas, did not necessarily imply ignorance, although it may have reduced their familiarity with particular local contexts. South Africa and Zimbabwe were sub-metropolises for southern and central Africa, and beyond. The botanist and ecologist, John Phillips, for instance, worked in the Knysna Forest in the Cape before assisting with the anti-tsetse campaigns in Tanzania during the 1930s. Theiler and du Toit carried out epidemiological surveys in other parts of the continent and advised the British government

24. Gilfoyle, *Veterinary Science*; Karen Brown, 'Tropical medicine and animal diseases: Onderstepoort and the development of veterinary science in South Africa, 1908–1950', *Journal of Southern African Studies* 31, 3 (2005), pp. 513–29.

on disease control policies in a number of its colonies.²⁵ The Indian civil service also supplied scientific officials to Africa, for example irrigation engineers Francis Kanthack in southern Africa and William Willcocks on the Nile. Kanthack worked in the Punjab before becoming Cape and Union Director of Irrigation (1906–23), overseeing the biggest dam projects in southern Africa, and then consulting in British colonies such as Malawi. A number of forestry officers followed similar routes, such as Ernest Hutchins who worked in Madras, and then spent 24 years in the Cape before becoming the Chief Conservator of Forests in British East Africa and Uganda in 1906.²⁶

A few also moved from southern Africa to Asia and back. Theiler appointed the Dublin-trained vet David Mitchell and detailed him to investigate *lamsiekte*, later identified as a form of botulism, which was linked to mineral deficiencies in pasture in the northern Cape. This research was of considerable significance for animal husbandry both in South Africa and in other parts of Africa with similar pasture profiles. In 1928, Mitchell was appointed Director of Veterinary Services for Burma. He investigated the problem of anthrax in elephants, using vaccines developed at Onderstepoort, and also improved rinderpest prophylaxis. In 1940, he headed a field laboratory as part of a campaign in Tanzania to suppress rinderpest. He returned to South Africa to work on typhus vaccines. In a number of publications and reports, he contributed to a growing body of knowledge about animal nutrition and the control of animal diseases.²⁷

Such scientific officials also drew on a wider range of experimentation and information than they could generate from their own research or from their particular colonial contexts. They shared information through correspondence, scientific journals, conferences, and government agricultural publications. Britain, which remained the major site of training for South African vets until the 1920s, Europe, North America, and Australia were important points of reference.²⁸ Knowledge acquired and techniques

25. John Phillips, 'The application of ecological research methods to the tsetse (*Glossina* spp) problem in Tanganyika territory: a preliminary account', *Ecology* 11, 4 (1930), pp. 713–33; Thelma Gutsche, *There Was a Man: The life and times of Sir Arnold Theiler KCMG of Onderstepoort* (Howard Timmins, Cape Town, 1979); Karen Brown, 'A sub-imperial science? South African veterinary medicine, the metropole and the wider world 1900–1950' (paper presented at the Commonwealth History Workshop, 'Science and Empire', Oxford, 12 May 2006).

26. Karen Brown, 'The conservation and utilisation of the natural world: silviculture in the Cape Colony c. 1902–1910', *Environment and History* 7, 4 (2001), pp. 427–47.

27. Daniel Gilfoyle, 'South Africans abroad: science and sub-imperialism in the control of rinderpest in Tanganyika, 1938–42' (paper presented at the Wellcome Unit Seminar, Oxford, 2005).

28. Thomas Dunlap, *Nature and the English Diaspora: Environment and history in the United States, Canada, Australia and New Zealand* (Cambridge University Press, Cambridge, 1999); Ian Tyrell, *True Gardens of the Gods: Californian-Australian environmental policy 1860–1930* (University of California Press, Berkeley, CA, 1999).

developed in these other regions were sometimes absorbed and incorporated into the work of experts in Africa. The study of the tick-borne cattle disease Texas fever by Theodore Smith and Francis Kilborne, for example, carried out in the United States in the early 1890s, proved of enormous significance for the control of animal diseases in Africa.²⁹

The first official research in Africa into tick-borne diseases – which were shown to be major killers – soon followed in the Cape Colony, after the appointment of the continent's first professional entomologist in 1895. Charles Lounsbury, trained in the United States, worked in South Africa until his retirement in 1927. He accepted the post, despite low pay, because there appeared to be so many animal and plant diseases that had never been studied in Africa and he longed for the international renown that could accrue from important scientific discoveries. Lounsbury and his veterinary colleagues demonstrated that a range of infections, including the devastating cattle disease East Coast fever, were conveyed by ticks. This paved the way for state-organized dipping campaigns to control such diseases.³⁰ Vets could be imperious in enforcing the dipping regulations and certainly provoked opposition.³¹ Nonetheless, if it is a legitimate function of the state to mitigate virulent diseases, South Africa's veterinary department had some success. By the 1920s, East Coast fever was largely eradicated and outbreaks of other tick-borne infections had sharply declined in number, to the benefit of both black and white livestock owners. While the specific way in which dipping was implemented, and the fines for breaching regulations, were often contested, the practice was gradually embedded in the rhythms of rural life.

What strikes us at the end of a sequence of projects on colonial veterinary and environmental history is not the ignorance of colonial scientists, but how much they came to understand and how quickly. It is not the brevity of their sojourns but the number of examples of sustained and varied engagement. They gained striking insights into animal diseases, entomology, and parasitology in a fertile period of global scientific work that also revealed the causes of malaria and trypanosomosis.³² Scientists working in the colonies

29. Theodore Smith and Francis Kilborne, *Investigations into the Nature, Causation and Prevention of Southern Cattle Fever* (Government Printing Office, Washington, 1893).

30. Paul Cranefield, *Science and Empire: East Coast fever in Rhodesia and the Transvaal* (Cambridge University Press, Cambridge, 1991); Daniel Gilfoyle, 'The heartwater mystery: veterinary and popular ideas about tick-borne animal diseases at the Cape, c. 1877–1910', *Kronos* 29 (2003), pp. 139–60; Karen Brown, 'Political entomology: the insectile challenge to agricultural development in the Cape Colony, 1895–1910', *Journal of Southern African Studies* 29, 2 (2003), pp. 529–49.

31. Colin Bundy, "'We don't want your rain and we won't dip': popular opposition, collaboration and social control in the anti-dipping movement, 1908–16' in Beinart and Bundy, *Hidden Struggles*, pp. 222–69.

32. Michael Worboys, 'Germs, malaria and the invention of Mansonian medicine: from "diseases in the tropics" to "tropical diseases"' in D. Arnold (ed.), *Warm Climates and Western Medicine* (Rodopi, Amsterdam, 1996), pp. 181–207.

contributed to Charles Wenyon's wide-ranging reference book on the subject, published in 1926.³³ The first southern African veterinary text books predated this, and, by the 1930s, specialist texts followed.³⁴ Much of the work tried to be interdisciplinary, as we now wish to be, pushing the frontiers of knowledge in a number of spheres and attempting to bridge both field and laboratory work.

Experts' reports, read, as it were, through their own eyes, for their own logic, rather than simply for their mistakes, or for the origins of authoritarian planning, or for opposition to them, become fascinating documents. Scientific reports also help to reveal how preliminary understandings evolved, and they are still very useful for non-specialists because they were often designed for an administrative or public readership. It is easy to underestimate the centrality of their scientific language, and conceptualizations, in shaping our basic grasp of African ecologies, natural history, and diseases – the ideas and terminology with which most historians unselfconsciously work. There are many environmental explanations and relationships, which are now common currency (even if they are not always entirely correct), that have their origins in these scientific networks.

Those who generated knowledge were not necessarily particularly talented. From the late nineteenth century, southern African problems certainly attracted some prominent international scientists, such as Robert Koch, who investigated rinderpest and tick-borne diseases, and David Bruce, who worked on trypanosomosis. Koch's relatively brief African and Indian sojourns epitomized high-profile interventions where problems were misdiagnosed and prescriptions of limited value.³⁵ Many of those who worked in the colonies may have been amongst the less successful, professionally speaking. They were able to make significant advances because they were trained in generalizing and comparative ways of thinking that enabled them to draw on discoveries elsewhere, discern, isolate, categorize, and begin to explain highly complex phenomena. It should also be acknowledged that some failed outright. Alexander Edington, a Scottish medical doctor and bacteriologist, was appointed in 1891 to direct the Cape's Bacteriological Institute and to investigate animal diseases. The Cape government

33. Charles Wenyon, *Protozoology: A manual for medical men, veterinarians and zoologists* (Baillière, Tindall, and Cox, London, 1926).

34. Charles Edmonds, *Diseases of Animals in South Africa* (Baillière, Tindall, and Cox, London, 1922); H. O. Mönnig, *Veterinary Helminthology and Entomology: The diseases of domesticated animals caused by helminth and arthropod parasites* (Baillière, Tindall, and Cox, London, 1934); D. G. Steyn, *The Toxicology of Plants in South Africa together with a Consideration of Poisonous Foodstuffs and Fungi* (Central News Agency, Johannesburg, 1934).

35. Cranefield, *Science and Empire*; Daniel Gilfoyle, 'Veterinary research and the African rinderpest epizootic: the Cape Colony 1896–98', *Journal of Southern African Studies* 29, 1 (2003), pp. 133–54.

closed the Institute in 1905, Edington having failed to offer much in the way of solutions.³⁶

In the light of these points, the category of colonial science, which Mark Harrison has recently described as ‘little more than a label of convenience, lacking precise definition and of questionable utility’, should be interrogated further.³⁷ The literature on the shortcomings of colonial science in Africa, or its exploitative character, tends to link scientific work closely with the social and political aims of colonial states or particular capitalist interests – such as plantation owners or the mining industry. With the exception of tropical medicine, however, accounts of the generation of scientific knowledge and ideas, as well as the practice of laboratory experiment and fieldwork, have been few. The historiography of colonial science has tended to be more concerned with its political nature rather than the activities of scientists and the history of scientific experiment and invention.³⁸

In this respect, the dominance of social constructivist approaches to colonial scientists begs the question of ‘which context?’ The context, for some of them, was not always nor only a single authoritarian colonial state, or capitalist enterprise serving the interests of the imperial centre and intent on controlling peasants. Many were working within rapidly evolving international disciplines, sometimes with fractious internal disputes. Their research agendas were certainly shaped to some degree by their institutional contexts. But their reference points were often wider. Their career paths traversed different places and could be linked to a variety of institutions as these became more central in international scientific work.³⁹ Some were also connected to broader developments through reading and publishing in British and international journals; in this way they maintained contact with global, and quite unpredictable, flows of information. As will be illustrated, local networks, practices, and discussions could also influence their work.

It may be more useful, as Roy MacLeod suggests, to view some experts as scientists in the colonies rather than colonial scientists.⁴⁰ To provide a more rounded analysis, we should explore how scientific knowledge has been, to varying extents, shaped by disciplinary dynamics and by the object of study, the natural world, as well as social and political forces.⁴¹ Or, to use

36. Gilfoyle, *Veterinary Science and Public Policy*; Ngqabutho Madida, *A History of the Colonial Bacteriological Institute, 1891–1905* (University of Cape Town, unpublished MA dissertation, 2003).

37. Mark Harrison, ‘Science and the British Empire’, *Isis* 96 (2005), p. 63.

38. Roy MacLeod, ‘Introduction’ to ‘Nature and empire: science and the colonial enterprise’, *Osiris* 15 (2000), pp. 1–13.

39. Paolo Palladino and Michael Worboys, ‘Science and imperialism’, *Isis* 83 (1993), pp. 91–102.

40. MacLeod, ‘Introduction’, pp. 1–13.

41. Michael Worboys, *Spreading Germs: Disease theories and medical practice in Britain, 1865–1900* (Cambridge University Press, Cambridge, 2000).

an old neo-Marxist idea developed to cope better with analysing ideologies and consciousness, scientific ideas and practices possessed a certain relative autonomy shaped also by scientific theory, debates, institutional politics, personal ambitions, knowledge, networks, and beliefs.

Knowledge, information flows, and unpredictability

Historians of science may find our emphasis on knowledge and institutions an obvious statement. The Africanist's riposte would be that some earlier histories of science in Africa were heroic, or too narrowly focused. We need to find a route that can cope both with a range of contexts and influences, as well as the specifics of scientific work in particular settings. Some of these considerations underlie shifts in theoretical writing on science in the colonies over the last thirty years. Roy MacLeod argued that although colonial scientific institutions sometimes attained a degree of autonomy from imperial control, they operated as part of a 'moving metropolis'.⁴² More recently, MacLeod, and especially David Wade-Chambers and Richard Gillespie, have characterized science in the colonies as part of the 'polycentric communications network' of modern science, with multidirectional information flows.⁴³ Similarly, in a post-colonial context, where the settings for scientific work have become ever more diverse, and the politics of knowledge central to new social movements for global justice, scientific ideas can be harnessed to a wide range of social and political projects with unpredictable results.

With respect to the diversity of networks in the period before 1960, Dan Gilfoyle has shown, for example, that veterinary work in southern Africa drew not only from Arnold Theiler's European contacts but also from those of his son Max, at the Rockefeller Institute in the United States, where he was rewarded with a Nobel prize for his work on yellow fever vaccines in the 1930s. The technique of using mice brains as a culture medium was transposed back to Onderstepoort to produce vaccines for African horse-sickness, amongst other animal diseases. As in the case of bluetongue in sheep, recently the focus of attention in a warming Europe, the disease is transmitted by midges of the genus *Culicoides*. South African research during the 1930s and 1940s became important in the history of veterinary immunology more generally, especially as these African diseases became more widely distributed through the world from the 1950s.⁴⁴

42. Roy MacLeod, 'On visiting the moving metropolis: reflections on the architecture of imperial science', *Historical Records of Australian Science* 5, 3 (1982), pp. 1–16.

43. David Wade-Chambers and Richard Gillespie, 'Locality in the history of colonial science', *Osiris* 15 (2000), pp. 221–40; Harrison, 'Science and the British Empire'.

44. Daniel Gilfoyle, 'Veterinary immunology as colonial science: method and quantification in the investigation of horsesickness in South Africa, c.1905–1945', *Journal of the History of Medicine and Allied Sciences* 61, 1 (2006), pp. 26–65.

Even in the case of longstanding European animal diseases, the flow of knowledge and technology was occasionally reversed. A case in point is Max Sterne's work on anthrax during the 1930s and 1940s. Sterne was appointed to supervise anthrax vaccine production at Onderstepoort and to research a means of improving the inoculation, which sometimes produced outbreaks of the disease.⁴⁵ By manipulating the atmosphere in which the bacterial cultures were grown, Sterne removed the capacity of the vaccine to spread disease, thus solving a problem that had bedevilled vaccination since its first demonstration by Louis Pasteur in the early 1880s. Sterne's invention spread throughout the world from the 1940s, replacing other techniques of immunization. It long remained the standard method of animal vaccination against anthrax and provided the basis for subsequent research into a human prophylaxis. Anthrax vaccination in South Africa also demonstrated the sometimes ambiguous relationship between colonial science and colonial subjects. In terms of numbers, African cattle owners were probably the major beneficiaries.⁴⁶ South African research and interventions, designed to control the deadly viral disease rinderpest, paved the way for the formation of the Pan-African Bureau of Epizootic Diseases, based in Kenya.⁴⁷ The organization and its successors, together with international aid organizations, played a key role in the large-scale vaccination programmes against contagious animal diseases, aimed at eradication, carried out across the Sahel since the 1960s.⁴⁸

As we have noted, McCracken's concerns about flawed policies based on inadequate scientific research reverberate through the literature. We have tried to illustrate a little of the diversity of veterinary science in South Africa, its ramifications in other parts of the continent, and its relative sophistication in an international context. We are also concerned that, on occasions, critical texts are unsure in their understanding of what the experts did. Waller and Homewood analysed the historical encounter between veterinary scientists and Maasai pastoralists in Kenya and Tanzania.⁴⁹ They drew many interesting contrasts between Maasai and Western understandings of disease and the respective means of disease management.

But Waller and Homewood tell us little about the historic development of Western veterinary science in a context in which practitioners could not

45. Rudolph Bigalke, 'The fourteen editors of the *Journal of the South African Veterinary Association*, 1927–2000', *Journal of the South African Veterinary Association* 71, 2 (2000), pp. 68–76.

46. Daniel Gilfoyle, 'Anthrax vaccination in South Africa: economics, experiment and the mass vaccination of animals, c. 1900–1945', *Medical History* 50, 4 (2006), pp. 465–90.

47. Gilfoyle, 'South Africans abroad'.

48. Clive Spinage, *Rinderpest: A history* (Kluwer Academic/Plenum Publishers, New York, NY, 2003).

49. Richard Waller and Katherine Homewood, 'Elders and experts: contesting veterinary knowledge in a pastoral community' in A. Cunningham and B. Andrews (eds), *Western Medicine as Contested Knowledge* (Manchester University Press, Manchester, 1997), pp. 69–93.

readily turn to metropolitan answers. They discussed malignant catarrhal fever, a dangerous viral disease for cattle, contracted from pastures shared with calving wildebeest. Maasai pastoralists were aware that the disease was connected with wildebeest and avoided pasture where they had been grazing. Waller and Homewood concluded that the practice of the Maasai is 'based on a complex system of external symptomology and causation', which enables them to identify the disease and indicate therapy. In contrast, veterinary medicine was based on a method which 'located cause in an unseen world of pathogens and separated cause and symptom in its classificatory system'.⁵⁰ Certainly vets would have thought of the disease as 'caused' by a micro-organism, according to the principles and practices of microbiology. But vets also judged the presence of disease by the observation of external symptoms, simple diagnostic technology such as the thermometer, and post-mortem lesions. For vets environmental explanations were at times important in relation to particular diseases and they were equally concerned to know that malignant catarrhal fever was 'caused' by contact with material deposited by calving wildebeest. There are many examples of vets drawing on local observations and attempting to test these.

In summary, we suggest that while the origins of particular patterns of knowledge are always important, it is problematic to see scientific knowledge and applied technical practices as static social artefacts, trapped in the context in which they were generated. This may be a difficulty with constructivist analysis more generally, in that the social contexts through which ideas pass can change. This point may apply particularly to the discipline of ecology. We know that significant developments in the natural sciences and environmental thinking emerged at the peripheries of the European imperial world and Libby Robin, amongst others, has argued that ecology was, on an international scale, partly a 'science of settlement' and a 'science of empire'.⁵¹ Peder Anker firmly located ecology as a discipline of imperial exploitation.⁵² However, while ecology became hitched to a number of different intellectual agendas in the twentieth century, a major thrust has been as a totalizing, intellectually adventurous discipline. And it could also be, as Helen Tilley suggests, highly subversive.⁵³ It was subversive of disciplinary boundaries, and it focused on complex causes and

50. Waller and Homewood, 'Elders and experts,' pp. 78–9.

51. Libby Robin, 'Ecology: a science of empire' in Tom Griffiths and Libby Robin (eds), *Ecology and Empire: Environmental history of settler societies* (Keele University Press, Edinburgh, 1997), pp. 63–75; Richard Grove, *Green Imperialism: Colonial expansion, tropical island Edens and the origins of environmentalism 1600–1860* (Cambridge University Press, Cambridge, 1995).

52. Peder Anker, *Imperial Ecology: Environmental order in the British Empire, 1895–1945* (Harvard University Press, Cambridge, MA, 2001), p. 77.

53. Helen Tilley, *Africa as a 'Living Laboratory'*, p. 128, quoting Paul B. Sears. Helen Tilley, 'African environments and environmental sciences: the African Research Survey, ecological paradigms and British colonial development' in William Beinart and JoAnn McGregor (eds), *Social History of African Environments* (James Currey, Oxford, 2003), pp. 109–30.

environmental impacts. It introduced – potentially at least – the human factor into natural sciences, and provided one of the key bases for the modern environmental movement. The careers of some British ecologists traversed the colonial and post-colonial era, and they became environmental advocates in new institutional contexts. Ecology was surely a version of ‘foxism’ rather than ‘hedghogism’ and has shaped intellectual approaches far more broadly. We are all ecologists now, perhaps, and our thinking about global environmental change is at least partly rooted in the aspirations to panoptical vision once pursued by scientists in the colonies.

The interpenetration of knowledge

Evidence is accumulating not only of how much was understood by scientists in the colonies, but also of a degree of openness to local knowledge. This was another element of the context in which experts operated. We do not wish to overemphasize this point or to re-invert the argument again. As noted above, the training of African scientists, who may have brought local experience further into the mainstream, was delayed by racial barriers. But we suggest that branches of scientific enquiry that required the development of expertise in the field in Africa, were not a closed system of ideas and practices, stuck in the laboratory. Fieldwork or observation often necessitated interaction with African people as well as observation of colonized nature, and, metaphorically speaking, the laboratory could have multiple locations.

Imbrications of knowledge have perhaps been more fully illustrated in the historiography of South Asia.⁵⁴ In the eighteenth century, some British officials, as ‘orientalists’, were deeply absorbed in recording Indian achievements. Christopher Bayly comments that ‘colonial knowledge was derived to a considerable extent from indigenous knowledge, albeit torn out of context and distorted by fear and prejudice’.⁵⁵ The literature, however, suggests a growing gap in the nineteenth century, so that India was increasingly seen not so much for its technological legacy as for its spirituality; British knowledge was seen as essential for progress.⁵⁶ Clearly the degree of openness of scientific research, and the scale of interpenetration of knowledge, changed through time and place, and differed across disciplines. In the African context, we can see such processes at work in the texts of late-eighteenth-century scientific travellers at the Cape, but also in the inter-war years of the twentieth century in British-controlled colonies.⁵⁷

54. Adas, *Machines as the Measure of Men*, p. 99.

55. Christopher Bayly, *Empire and Information: Intelligence gathering and social communication in India, 1780–1870* (Cambridge University Press, Cambridge, 1996), p. 7.

56. Satpal Sangwan, *Science, Technology and Colonisation: An Indian experience, 1757–1857* (Anamika Prakashan, Delhi, 1991); MacLeod and Kumar, *Technology and the Raj*.

57. Beinart, ‘Men, science’; Tilley, ‘African environments and environmental sciences’.

McCracken himself cites colonial officers learning from Africans about cotton planting times. William Willcocks, one of the British Empire's leading dam builders, was by the 1910s and 1920s extolling the virtues of traditional Egyptian irrigation systems.⁵⁸ The Zambian ecological and agricultural research of Colin Trapnell and William Allan, from the 1930s to the 1950s, is well known for both its rigour in the context of that time and its sensitivity to local systems of agriculture.⁵⁹ Edgar Barton Worthington, a key figure in imperial ecology, who wrote the volume on science for Lord Hailey's monumental *African Survey* (1938), cut his teeth studying the fisheries of Lake Victoria during the depression years.⁶⁰ His work was intended to understand the ecology of the lake at a time when overfishing was seen to be threatening food supplies. He openly admitted that he learnt more from African fishermen than they did from him.

There was also a more diffuse pattern of incorporating African names for species and diseases which reflects a longer history of learning and interaction, often by travellers and settlers as much as officials and scientists. Tsetse was taken from the Setswana name for this fly. Nagana, the commonly used term for bovine trypanosomosis, is the anglicization of *uNakane*, a Zulu word meaning depressed or low in spirits, which describes the bedraggled and forlorn aspect of sick animals. David Bruce used this word when reporting his research in Zululand during the late 1890s and it spread throughout the anglophone colonial world. In South Africa, old established Dutch names for animals, plants, and diseases, some in turn drawn from the Khoisan, were incorporated into scientific as well as popular language. *Gifblaar*, or 'poison leaf', which kills if ingested, *lamsiekte* (lame sickness, as noted above a symptom of botulism), *vermeersiekte* (vomiting sickness) all remained in common scientific usage, describing the symptoms rather than the names of the offending chemical or micro-organism. Sometimes this nomenclatural adoption was more deliberate: Cicely Williams deployed an indigenous Ghanaian category for illness, kwashiorkor, in identification of child malnutrition in the 1920s, and it also became universalized.⁶¹

At times African and settler knowledge made an important contribution to developments in the natural sciences, especially in the nineteenth century, when many disciplines were in their infancy. Karen Brown has argued that biomedical researchers drew on African ideas about the cause of

58. William Beinart and Lotte Hughes, *Environment and Empire* (Oxford University Press, Oxford, 2007).

59. William Allan, *The African Husbandman* (with a new introduction by Helen Tilley) (International Africa Institute, LIT Verlag, 2005).

60. Tilley, *Africa as a 'Living Laboratory'*, p. 88; Edgar Barton Worthington, *Science in Africa* (Oxford University Press, London, 1938); Edgar Barton Worthington, *The Ecological Century: A personal appraisal* (Oxford University Press, Oxford, 1983).

61. Cicely D. Williams, 'Kwashiorkor. A nutritional disease of children associated with a maize diet', *Lancet*, 16 November 1935, p. 1151.

nagana and its control.⁶² Zulu people identified links between cattle, tsetse fly, and wildlife. Some Zulu associated the disease with the contamination of the veld by the saliva of game, whilst others ascribed it to the presence of tsetse fly belts. When Bruce published a report on his research at Ubombo in northern Zululand in 1895, he described and expanded upon an aetiology that resonated with local observations of the disease. Zulu kings, and the successor states, had also tried to drive back tsetse infestation by clearing land of wildlife, and this became a frequently adopted colonial policy. Charles Swynnerton recorded how the chiefs in the Gaza kingdom in Mozambique located their subjects in closer settlements from which wildlife, as well as the thickets that provided shelter for the fly, were removed. Swynnerton adapted these methods in Tanzania during the 1920s and 1930s and claimed some success in containing the disease through villagization.⁶³ The work of both Bruce and Swynnerton demonstrated a close interaction between metropolitan science and African observations. However, the use of coercive concentrated settlements by colonial states did trigger resistance, and has also been a major focus of the critique of colonial science.⁶⁴

More generally, white and black livestock owners had a knowledge of toxic flora and patterns of transhumance were sometime pursued to avoid poisoning.⁶⁵ In certain respects, scientific work closed doors from the late nineteenth century, when germ theories and laboratory techniques became more central, and veterinary science became securely established as a discipline.⁶⁶ Livestock owners nonetheless continued to provide the animals and farms for field experiments and their observations remained important for monitoring the distribution of diseases. Ethnoveterinary research by scientists as well as anthropologists gradually expanded in the post-colonial period, and since 1994 Onderstepoort, Fort Hare, and other South African institutions have specifically promoted research on local knowledge in this field.

The two decades after the Second World War have been seen as the apex of interventionism and scientific hubris, the era of 'high modernism' in Scott's terms. Yet Grace Carswell has illustrated how colonial officials worked with chiefs to implement soil conservation measures in Uganda, modifying

62. Karen Brown, 'From Ubombo to Mkhuzi: disease, colonial science, and the control of Nagana (Livestock Trypanosomosis) in Zululand, South Africa, c. 1894–1953', *Journal of the History of Medicine and Allied Sciences Advance* 63, 3 (2008), pp. 285–322.

63. Charles Swynnerton, 'An examination of the tsetse problem in North Mossurise Portuguese East Africa', *Bulletin of Entomological Research* 11, 4 (1921), pp. 315–85; 'Tsetse flies of East Africa: a first study of their ecology with a view to their control', *Transactions of the Royal Entomological Society of London* 84 (1936), pp. 1–579.

64. Hoppe, *Lords of the Fly*; Lyons, *The Colonial Disease*.

65. Karen Brown, 'Poisonous plants, pastoral knowledge and perceptions of environmental change in South Africa, c. 1880–1940', *Environment and History* 13, 3 (2007), pp. 307–32; William Beinart, 'Transhumance, animal diseases and environment in the Cape, South Africa', *South African Historical Journal* 58 (2007), pp. 17–41.

66. Gilfoyle, *Veterinary Science and Public Policy*.

and incorporating local practice in implementation.⁶⁷ The malaria research community was deeply split, from the 1930s to the 1950s, between those who saw the disease as eradicable, and those, aware of the importance of sustaining local immunity, who wished to adopt a more limited approach.⁶⁸ This prefigured arguments about tsetse control. Ecologists such as Barton Worthington and Raymond Dassman developed the idea that wildlife farming would be ecologically beneficial, as well as economically profitable. Experiments were pursued in co-operation with Maasai district councils in Kenya as well as with settler farmers in southern Africa in the 1950s and 1960s.⁶⁹

John Ford's 1971 book about tsetse and trypanosomosis has been highly influential in Africanist debates as a major critique of colonial scientific objectives.⁷⁰ But he was a scientist in the colonies, and his career illustrates some of the points explored here. He trained at Oxford and went on one of the University's colonial expeditions – to Borneo. He made a sustained career in Africa, working in at least five locations. His work reflected debates and changing views within professional and official communities – and, as he tells us, his book 'had its origin in a conversation between its author and some Africans in Western Uganda in 1945'.⁷¹ He displayed an increasing understanding of local knowledge and sensitivity to the impact of interventions, advocating that prophylaxis should not be based on attempts at disruptive total eradication. Rather he was interested in exploring locally achieved balances between settlement, livestock keeping, and environmental control, and the possibility that this facilitated partial immunity to trypanosomosis.

Ford's work strongly influenced Helge Kjekshus, who wrote a foundational text in radical African environmental history.⁷² Kjekshus argued that colonialism had upset the ecological balance between people and environment in Tanzania, greatly facilitating a debilitating spread of tsetse fly. While his analysis did not replicate Ford's, there is a direct intellectual continuity between the approach of a scientist in the colonies and the critique of science and colonialism. It is an irony that Ford's approach may have

67. Grace Carswell, 'Soil conservation policies in colonial Kigezi, Uganda: successful implementation and an absence of resistance' in Beinart and McGregor, *Social History and African Environments*.

68. Mary Dobson and Maureen Malowany, 'DDT and malaria control in East Africa, 1945–1960: discoveries, debates and dilemmas' (unpublished paper, African Studies Seminar, St Antony's College, University of Oxford, 2000).

69. Dawn Nell, *The Development of Wildlife Utilization in South Africa and Kenya, c. 1950–1990* (University of Oxford, unpublished DPhil thesis, 2003).

70. John Ford, *The Role of the Trypanosomiases in African Ecology: A study of the tsetse fly problem* (Oxford University Press, London, 1971).

71. *Ibid.*, p. 7.

72. Helge Kjekshus, *Ecology Control and Economic Development in East African History* (Heinemann, London, 1977); James Giblin, 'East Coast fever in socio-historical context: a case study from Tanzania', *International Journal of African Historical Studies* 23, 3 (1990), pp. 401–21.

been misplaced when he eventually published his book in 1971. Systematic medical and environmental intervention in the late colonial and early independence years, together with economic processes such as the spread of cash cropping, land clearance and denser settlement, almost certainly reduced human trypanosomosis. Reported cases had declined to about 10,000 annually by the late 1960s. They increased again to a reported 45,000 deaths annually, probably more, by 1990.⁷³ The collapse of both medical services and the capacity to deal with the disease in the face of economic stress and social conflict were key factors.

Conclusion

Scientific and technical imaginations were brought to bear in securing conquest and enhancing production but also, for example, in seeing river valleys and watersheds as a whole, incorporating forestry, soil science, hydraulics, climatology, and ecology into planning. Even if scientists were often insensitive to local communities, their approach was not always, in Scott's terms, 'hedgehogism', but came closer to 'foxism' – viewing the landscape in its entirety, understanding multifaceted patterns of causation, and making a multitude of calculations about its potential. Our evidence suggests that some vets also had a regional overview of the causation and treatment of animal diseases that was not easy for non-specialists to attain. It would be very difficult to apply Scott's dictum on the technical officer or McCracken's on peasant knowledge to the understanding and prevention of regional animal diseases, or for that matter HIV/AIDS or climate change. We do not intend to make a cheap point here, and fully recognize that there are different types and levels of environmental problems, and diseases.

Experts also adopted complex models through which to understand and represent the world. On the one hand scientists were becoming more specialized in the growing array of disciplines. They often wrote on restricted topics, because the norms of discipline and publication demanded that they did so – historical research is hardly different. On the other hand, some perceived their contributions as part of a greater research effort and the development of general propositions. Veterinary scientists engaged in more general debates about public policy, environmental change, and conservation. Ecology, in particular, was a means of pushing the boundaries of interdisciplinary scientific enquiry. Scientists also increasingly shaped the concepts through which other disciplines and popular literature understood the natural environment and disease. Approaches to science were not static. There is evidence of an interpenetration of scientific and local knowledge, whether acknowledged or not.

73. World Health Organization, 'Tropical diseases, including Pan African tsetse and trypanosomiasis eradication campaign' (Fifty-Sixth World Health Assembly A56/9, Report by the Secretariat). <who.int/gb/ebwha/pdf_files/WHA56/> (19 January 2009).

To return to McCracken's points, African polities, colonial officials, independent states, and international agencies have all floundered in trying to deal with trypanosomosis and other intractable diseases. In some of the environmental history critique, there is almost a suggestion that these complex problems of control were solvable. In effect, much colonial intervention was experimental and outcomes were unpredictable. As McCracken illustrated, the tsetse fly in Malawi was in part driven back largely by expansion of tobacco cash cropping and penetration of market relations from the 1930s.⁷⁴ In the post-war era, scientists in South Africa used DDT to eradicate *Glossina pallidipes* from Zululand.⁷⁵ States and international organizations achieved a considerable degree of control over the human forms of the disease by the early independence period. In South Africa, where dipping was universally enforced, it did prove possible to eliminate East Coast fever and contain other tick-borne diseases. State veterinary regulation in South Africa is now ebbing, and African livestock owners are bearing the consequences.

The study of African knowledge and responses remains essential.⁷⁶ Nor should we dodge difficult questions about colonialism or top-down and coercive planning. There is no doubt that agricultural and conservationist interventions provoked opposition and helped to fuel wider anti-colonial political movements. This must remain a major framework for historical interpretation. But our article asks whether Africanist literature has become trapped in a critique of science and whether it obscures interesting and important questions about scientific and technical ideas that have provided the building blocks for understanding environment and disease in Africa. We have suggested that the sites for scientific work, networks, and protagonism became more diverse, with unpredictable outcomes. This is even more the case in the post-colonial context. The debates over HIV/AIDS treatment in South Africa are an important example, which have influenced our views. There, radical AIDS activists in the Treatment Action Campaign have engaged in a new politics of knowledge, mobilizing scientific arguments in opposition to Mbeki's Africanist critique of science that stalled effective government intervention. Our article seeks routes by which to explore further the salience and fascination of scientific knowledge, and its value in policy and practice. The increasing flexibility of scientific thinking certainly helps in this respect.

74. John McCracken, 'Colonialism, capitalism and ecological crisis in Malawi: a reassessment' in David Anderson and Richard Grove (eds), *Conservation in Africa: People, policies and practice* (Cambridge University Press, Cambridge, 1987).

75. Brown, 'From Ubombo to Mkhuzi'.

76. And is the subject of an ESRC-sponsored research project on veterinary history by Brown and Beinart.